

Data Programming in Python | BAIS:6040

Homework 1 Solution

Strings

Download and open declaration.txt and constitution.txt. Copy and paste the text inside a multi-line string in Python

```
In [20]: """
Multiline comment
To set a multiline string object use the three quotation marks like in this cell.
"""

comment = """
Will this work?
I really hope so!
"""
```

```
In [ ]:
```

```
In [ ]: print(comment)
```

```
In [3]: comment
```

```
Out[3]: '\nWill this work?\nI really hope so!\n'
```

```
In [4]: declaration = """
In Congress, July 4, 1776

The unanimous Declaration of the thirteen united States of America, When in the Cou

We hold these truths to be self-evident, that all men are created equal, that they

He has refused his Assent to Laws, the most wholesome and necessary for the public

He has forbidden his Governors to pass Laws of immediate and pressing importance, u

He has refused to pass other Laws for the accommodation of large districts of peopl

He has called together legislative bodies at places unusual, uncomfortable, and dis

He has dissolved Representative Houses repeatedly, for opposing with manly firmness

He has refused for a long time, after such dissolutions, to cause others to be elec

He has endeavoured to prevent the population of these States; for that purpose obst

He has obstructed the Administration of Justice, by refusing his Assent to Laws for
```

He has made Judges dependent on his Will alone, for the tenure of their offices, and the value of their Decisions.

He has erected a multitude of New Offices, and sent hither swarms of Officers to harass and oppress the People.

He has kept among us, in times of peace, Standing Armies without the Consent of our Legislature.

He has affected to render the Military independent of and superior to the Civil power.

He has combined with others to subject us to a jurisdiction foreign to our Constitution, and unacknowledged by our Laws; For Quartering large bodies of armed troops among us:

For protecting them, by a mock Trial, from punishment for any Murders which they should have committed on the Spot.

For cutting off our Trade with all parts of the world:

For imposing Taxes on us without our Consent:

For depriving us in many cases, of the benefits of Trial by Jury:

For transporting us beyond Seas to be tried for pretended offences:

For abolishing the free System of English Laws in a neighbouring Province, establishing instead thereof a Tyrannical Government.

For taking away our Charters, abolishing our most valuable Laws, and altering fundamentally the Forms of our Governments.

For suspending our own Legislatures, and declaring themselves invested with power to enact Laws without us.

He has abdicated Government here, by declaring us out of his Protection and waging War against us.

He has plundered our seas, ravaged our Coasts, burnt our towns, and destroyed the lives of many thousands of our People.

He is at this time transporting large Armies of foreign Mercenaries to compleat the Execution of these atrocious Schemes.

He has constrained our fellow Citizens taken Captive on the high Seas to bear Arms against their Country.

He has excited domestic insurrections amongst us, and has endeavoured to bring on the general Civil War.

In every stage of these Oppressions We have Petitioned for Redress in the most humble Manner: but our Petitions have been always denied.

Nor have We been wanting in attentions to our Brittish brethren. We have warned the same by every opportunity.

We, therefore, the Representatives of the united States of America, in General Congress assembled, do hereby declare, that these United Colonies are, and of right ought to be, free and independent States; that they are absolved from all Allegiance to the British Crown, and that all future Interference by the British Crown with these States is prohibited, and that every State, united to them, or united to them by Treaty, Intercourse, or any other Means, shall have complete Power to do the same.

```
In [5]: constitution = """
We the People of the United States, in Order to form a more perfect Union, establish and define the Rights of the People, do hereby adopt this Constitution for the United States of America.

Article. I.
Section. 1.

All legislative Powers herein granted shall be vested in a Congress of the United States, which shall consist of a Senate and House of Representatives.

Section. 2.

The House of Representatives shall be composed of Members chosen every second Year
```

The Congress shall have Power To lay and collect Taxes, Duties, Imposts and Excises

To borrow Money on the credit of the United States;

To regulate Commerce with foreign Nations, and among the several States, and with t

To establish an uniform Rule of Naturalization, and uniform Laws on the subject of
 To coin Money, regulate the Value thereof, and of foreign Coin, and fix the Standar
 To provide for the Punishment of counterfeiting the Securities and current Coin of
 To establish Post Offices and post Roads;
 To promote the Progress of Science and useful Arts, by securing for limited Times t
 To constitute Tribunals inferior to the supreme Court;
 To define and punish Piracies and Felonies committed on the high Seas, and Offences
 To declare War, grant Letters of Marque and Reprisal, and make Rules concerning Cap
 To raise and support Armies, but no Appropriation of Money to that Use shall be for
 To provide and maintain a Navy;
 To make Rules for the Government and Regulation of the land and naval Forces;
 To provide for calling forth the Militia to execute the Laws of the Union, suppress
 To provide for organizing, arming, and disciplining, the Militia, and for governing
 To exercise exclusive Legislation in all Cases whatsoever, over such District (not
 To make all Laws which shall be necessary and proper for carrying into Execution th
 Section. 9.
 The Migration or Importation of such Persons as any of the States now existing shal
 The Privilege of the Writ of Habeas Corpus shall not be suspended, unless when in C
 No Bill of Attainder or ex post facto Law shall be passed.
 No Capitation, or other direct, Tax shall be laid, unless in Proportion to the Cens
 No Tax or Duty shall be laid on Articles exported from any State.
 No Preference shall be given by any Regulation of Commerce or Revenue to the Ports
 No Money shall be drawn from the Treasury, but in Consequence of Appropriations mad
 No Title of Nobility shall be granted by the United States: And no Person holding a
 Section. 10.
 No State shall enter into any Treaty, Alliance, or Confederation; grant Letters of
 No State shall, without the Consent of the Congress, lay any Imposts or Duties on I
 No State shall, without the Consent of Congress, lay any Duty of Tonnage, keep Troo
 Article. II.
 Section. 1.

The executive Power shall be vested in a President of the United States of America.

Each State shall appoint, in such Manner as the Legislature thereof may direct, a Number of Electors, who shall have the Qualifications and Powers, which the Legislature of each State may think proper to require.

The Electors shall meet in their respective States, and vote by Ballot for two Persons, one of whom shall be a natural born Citizen, or a Citizen of the United States, and who shall not be an Inhabitant of that State in which they meet.

The Congress may determine the Time of chusing the Electors, and the Day on which they shall assemble.

No Person except a natural born Citizen, or a Citizen of the United States, at the Time of chusing, shall be eligible to this Office.

In Case of the Removal of the President from Office, or of his Death, Resignation, or Inability to discharge the Duties of this Office, the same shall devolve on the Vice President, and in Case the Vice President shall also be removed from Office, or die, then the Electors shall chuse a new President, who shall act until the next Meeting of the Electors.

The President shall, at stated Times, receive for his Services, a Compensation, which shall neither be increased nor diminished during his Continuance in Office.

Before he enter on the Execution of his Office, he shall take the following Oath or Affirmation.

Section. 2.

The President shall be Commander in Chief of the Army and Navy of the United States; he shall have Power to grant Reprieves and Pardons for all Crimes and Offences, except in Cases of Impeachment;

He shall have Power, by and with the Advice and Consent of the Senate, to make Treaties, provided two thirds of the Senators present concur;

He shall have Power to fill up all Vacancies that may happen during his Continuance in Office, by and with the Advice and Consent of the Senate.

Section. 3.

He shall from time to time give to the Congress Information of the State of the Union, and recommend to their Consideration such Measures as he shall judge necessary and proper.

Section. 4.

The President, Vice President and all civil Officers of the United States, shall be elected for four Years; and no Person shall be elected to this Office more than twice, but any Person who has held the Office of President, or acted as President, for two Years, shall not be eligible for another Election.

Article. III.

Section. 1.

The judicial Power of the United States, shall be vested in one supreme Court, and in such inferior Courts as the Congress may from time to time ordain and establish.

Section. 2.

The judicial Power shall extend to all Cases, in Law and Equity, arising under this Constitution, the Laws of the United States, and Treaties made, or which shall be made, under the Authority of the United States;

In all Cases affecting Ambassadors, other public Ministers and Consuls, and those in which a State shall be Party, the supreme Court shall have original Jurisdiction.

The Trial of all Crimes, except in Cases of Impeachment, shall be by Jury; and such Trial shall be held in the State where the Crime shall have been committed.

Section. 3.

Treason against the United States, shall consist only in levying War against them, or in adhering to their Enemies, giving them Aid and Comfort.

The Congress shall have Power to declare the Punishment of Treason, but no Attainder of Treason shall be inflicted by the Congress, until after the Passage of a Bill of Attainder, by a two thirds Majority.

Article. IV.

Section. 1.

Full Faith and Credit shall be given in each State to the public Acts, Records, and judicial Proceedings, of every other State.

Section. 2.

The Citizens of each State shall be entitled to all Privileges and Immunities of Citizens of the United States.

A Person charged in any State with Treason, Felony, or other Crime, who shall flee from that State, so that he cannot be found, shall be apprehended by other States, and be removed to the State from which he fled, to be there punished according to the Laws of that State.

No Person held to Service or Labour in one State, under the Laws thereof, escaping into another, shall, in Consequence of any Law, Regulation, or Jurisdiction of the latter State, be discharged from such Service or Labour, but shall be delivered up to the Possession of the Owner to whom such Person shall have been lawfully sold.

Section. 3.

New States may be admitted by the Congress into this Union; but no new State shall

The Congress shall have Power to dispose of and make all needful Rules and Regulating

Section. 4.

The United States shall guarantee to every State in this Union a Republican Form of

Article. V.

The Congress, whenever two thirds of both Houses shall deem it necessary, shall propose

Article. VI.

All Debts contracted and Engagements entered into, before the Adoption of this Constitution

This Constitution, and the Laws of the United States which shall be made in Pursuance

The Senators and Representatives before mentioned, and the Members of the several State

Article. VII.

The Ratification of the Conventions of nine States, shall be sufficient for the Establishment

The Word, "the," being interlined between the seventh and eighth Lines of the first

Attest William Jackson Secretary

done in Convention by the Unanimous Consent of the States present the Seventeenth Day

""

Now use string inclusion to see which of these 3 words is not in the U.S. constitution

america democracy taxes

Remember to check both upper and lower case since Python is case sensitive

Do these checks in separate cells and after these checks use a Markdown cell to give the answer in a sentence.

```
In [6]: "america" in constitution
```

```
Out[6]: False
```

```
In [7]: "America" in constitution
```

```
Out[7]: True
```

```
In [8]: "democracy" in constitution
```

```
Out[8]: False
```

```
In [9]: "Democracy" in constitution
```

```
Out[9]: False
```

```
In [10]: "taxes" in constitution
```

```
Out[10]: False
```

```
In [11]: "Taxes" in constitution
```

```
Out[11]: True
```

Democracy is the word that is not in the constitution

Using the length built-in python function (len), save how many characters each document contains.

Then, see how much longer the Constitution is than the Declaration of Independence.

i.e. The Constitution is ____ times longer than the Declaration of Independence.

```
In [12]: constitution_len = len(constitution)
```

```
In [13]: declaration_len = len(declaration)
```

```
In [14]: print("The Constitution is", constitution_len/declaration_len, "times longer than t
```

The Constitution is 3.233836339099497 times longer than the Declaration of Independence.

Numbers

Visit <https://studentaid.gov/understand-aid/types/loans/interest-rates>

You are a graduate Iowa MSBA student taking out \$20,000 in direct unsubsidized loans. It is the first month of interest accruing, and it has been 25 days since your last payment (aka your loan started).

How much interest has accrued?

```
In [15]: outstanding_principal = 20000
```

```
In [16]: # This is your daily interest rate. Assume 365 days in a year
interest_factor = .0794/365
```

```
In [17]: num_days = 25
```

```
In [18]: interest_accrued = (outstanding_principal * interest_factor) * num_days
```

```
In [19]: print("The total interest accrued is $", interest_accrued)
```

The total interest accrued is \$ 108.76712328767124